



Wensheng Zhang

随机信号分析

Random signal analysis

课程号: sd01234090-1 周五1-2节

振声苑W102, 通信21[1-3]班, 2022年秋

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第一章作业



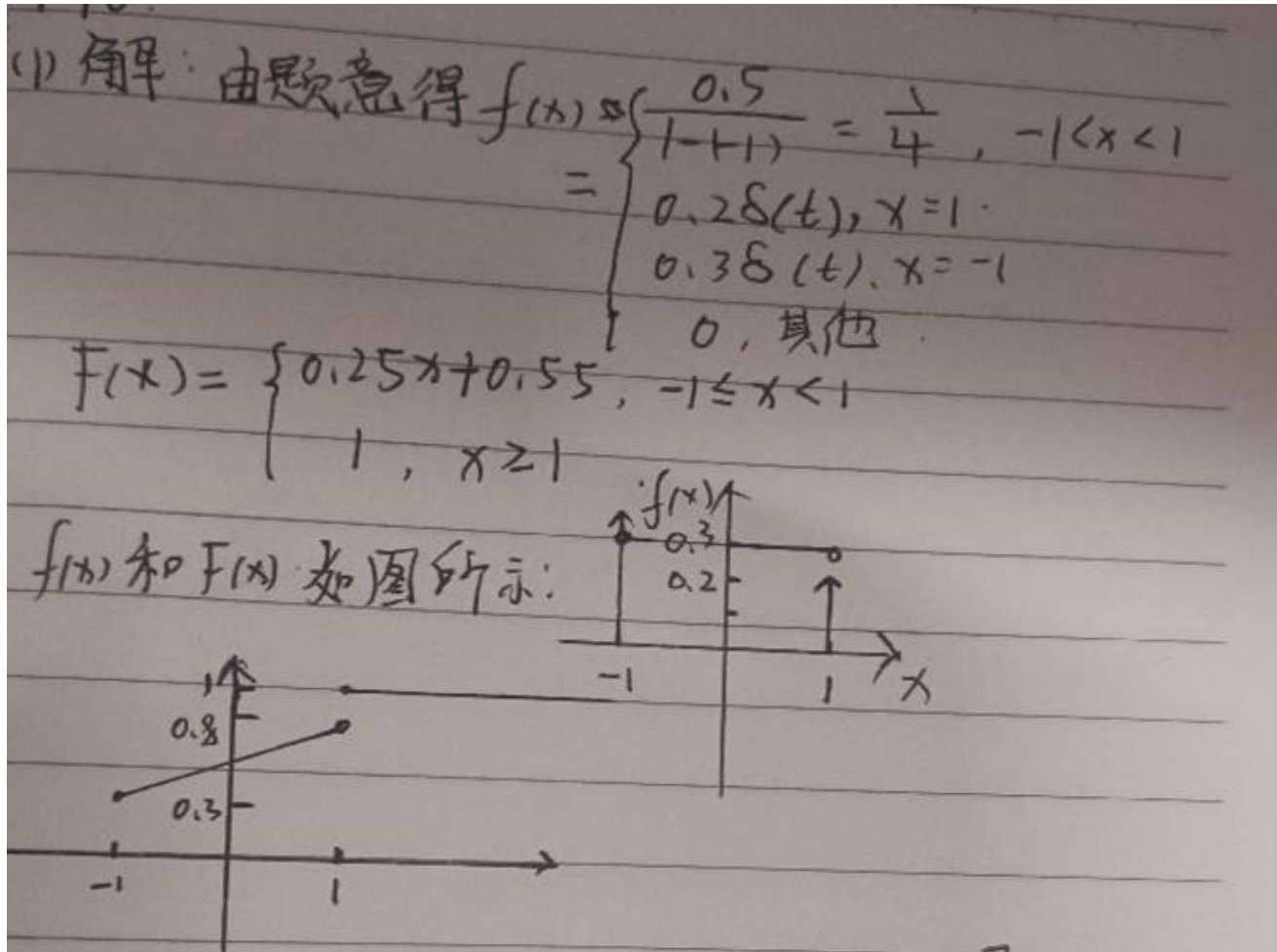
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$$1.6 \quad (1) \quad P(A_1) = P(A_2) = P(A_3) = P(A_4) = \frac{1}{4}$$

$$P(B) = 0.05 \times \frac{1}{4} + 0.4 \times \frac{1}{4} + 0.1 \times \frac{1}{4} + 0.1 \times \frac{1}{4} = 0.1625$$

$$\text{解} \quad P(A_2|B) = \frac{P(A_2)P(B|A_2)}{P(B)} = \frac{0.25 \times 0.4}{0.1625} = 0.615$$

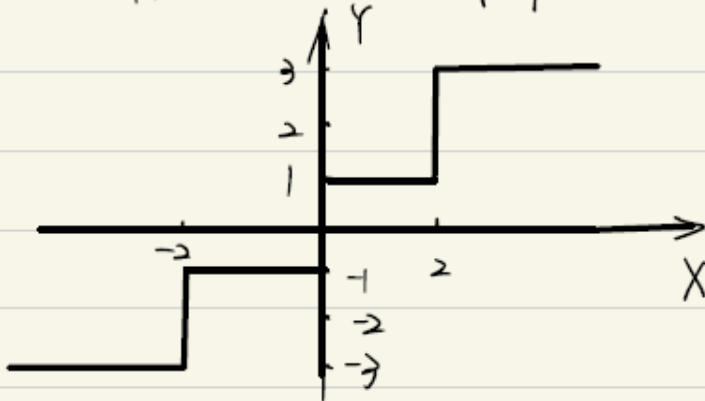
1.10 设随机变量 X 的绝对值不大于1。



1.11 分正负讨论 PDF全部面积为1

1.15 怎么算的？这么算对吗？ $N(\mu, \sigma^2)$ ？怎么算？

1.15 设高斯随机变量 $X \sim N(0, 2)$ 作用于一个 $L=4$ 电平的最优器，其量比特性如图 1.15 所示，试求输出随机变量 Y 的分布律。



Y 的取值一共有 4 个
分别为 $-3, -1, 1, 3$

$$p(Y=3) = p(X \geq 2) = \int_2^{\infty} \frac{1}{\sqrt{2\pi} \times 2} \exp\left(-\frac{x^2}{8}\right) dx = \int_2^{\infty} \frac{1}{\sqrt{2\pi} \times 2} e^{-\frac{x^2}{8}} = 0.1587$$

$$p(Y=1) = p(0 \leq X < 2) = 0.3413$$

同理得 $p(Y=-1) = 0.3413$

$$p(Y=3) = 0.1587$$

1.15. 解: $X \sim N(0, 2)$
 $Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$

$L=4 \therefore P\{Y=-3\} = P\{X \leq -2\} = P\{Z \leq -\sqrt{2}\} = 1 - \Phi(\sqrt{2}) = 0.0793$

$P\{Y=-1\} = P\{-2 < X \leq 0\} = 0.4207$

$P\{Y=1\} = 0.4207$

$P\{Y=3\} = 0.0793$

1.15

$P(Y=-3) = P(X \leq -2) = P\left(\frac{X}{\sqrt{2}} \leq \frac{-2}{\sqrt{2}}\right) = \Phi(-\sqrt{2}) \approx 0.0787$

$P(Y=-1) = P(-2 < X \leq 0) = P\left(\frac{-2}{\sqrt{2}} < \frac{X}{\sqrt{2}} \leq 0\right) = \Phi(0) - \Phi(-\sqrt{2}) \approx 0.4213$

$P(Y=1) = P(0 < X \leq 2) = P\left(0 < \frac{X}{\sqrt{2}} \leq \sqrt{2}\right) = \Phi(\sqrt{2}) - \Phi(0) \approx 0.4213$

$P(Y=3) = P(X > 2) = P\left(\frac{X}{\sqrt{2}} > \sqrt{2}\right) = 1 - \Phi(\sqrt{2}) \approx 0.0787$

1.16 证明独立。 相关、正交。 答案

1-16

$$E(U) = U_X + U_Y = 0 \quad E(V) = U_X - U_Y = 0$$

$$D(U) = \sigma_X^2 + \sigma_Y^2 = 1 + 1 = 2 \quad D(V) = \sigma_X^2 + \sigma_Y^2 = 1 + 1 = 2$$

$$\therefore U \sim N(0, 2) \quad V \sim N(0, 2)$$

$$\text{COV}(U, V) = E(UV) - E(U) \cdot E(V) = E(X^2 - Y^2) = 0$$

$$\rho_{UV} = \frac{\text{COV}(U, V)}{\sigma_U \sigma_V} = 0 \quad \therefore f_{U, V} = N(0, 2)$$

$$\therefore f_{UV}(u, v) = \frac{1}{4\pi} e^{-\frac{u^2 + v^2}{4}}$$

$$f_U(u) = \int_{-\infty}^{+\infty} \frac{1}{2\pi} e^{-\frac{u^2}{4}} \quad f_V(v) = \frac{1}{2\pi} e^{-\frac{v^2}{4}}$$

$$f_U(u) \cdot f_V(v) = \frac{1}{4\pi} e^{-\frac{u^2 + v^2}{4}} = f_{UV}(u, v)$$

$\therefore U, V$ 是相互独立的



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1.20 串联 并联 备用

20. (1) $f(x) = \frac{1}{\alpha} \alpha \cdot e^{-\alpha x}$, $f(y) = \beta \cdot e^{-\beta y}$, $Z = \min\{X, Y\}$

$$F_Z(z) = P\{Z \leq z\} = P\{\min(X, Y) \leq z\} = 1 - P\{\min(X, Y) > z\} = 1 - P\{X > z, Y > z\}$$

$$= 1 - [1 - F_X(z)][1 - F_Y(z)]$$

$$F_X(z) = \begin{cases} \int_0^z \alpha e^{-\alpha x} dx = 1 - e^{-\alpha z} & z \geq 0 \\ 0 & z < 0 \end{cases} \quad F_Y(z) = \begin{cases} 1 - e^{-\beta z} & z \geq 0 \\ 0 & z < 0 \end{cases}$$

1.20 ① $f_Z(z) = \begin{cases} (\alpha + \beta) e^{-(\alpha + \beta)z} & z \geq 0 \\ 0 & \text{其他} \end{cases}$

② $f_Z(z) = \begin{cases} \alpha e^{-\alpha z} + \beta e^{-\beta z} - (\alpha + \beta) e^{-(\alpha + \beta)z} & z \geq 0 \\ 0 & \text{其他} \end{cases}$

③ $f_Z(z) = f_X(z) * f_Y(z) = \begin{cases} \frac{\alpha \beta}{\beta - \alpha} (e^{-\alpha z} - e^{-\beta z}) u(z) & \alpha \neq \beta \\ \alpha \beta z e^{-\alpha z} u(z) & \alpha = \beta \end{cases}$

1.22 独立 相关 正交

一、‘独立’，‘不相关’和‘正交’的定义

三者定义：

对于两个随机变量X和Y，

(1) 独立： $p(x,y) = p_1(x) * p_2(y)$

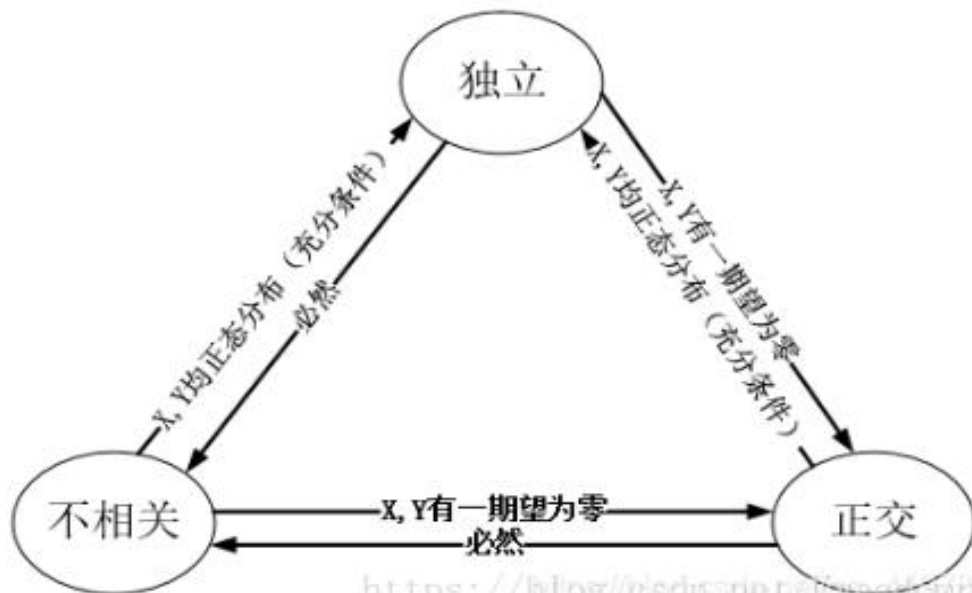
(2) 不相关： $Cov(X,Y) = 0$

$\Leftrightarrow E(X*Y) - E(X)*E(Y) = 0, E(X*Y) = E(X)*E(Y)$

(3) 正交： $E(X*Y) = 0$

<https://blog.csdn.net/Freedomroad>

二、“独立”、“不相关”和“正交”之间的关系



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1.29

$$(1) f(x) = 0.2\delta(x) + 0.3\delta(x-2) + 0.2\delta(x-4) + 0.2\delta(x+2) + 0.1\delta(x+4)$$

$$E(x) = 0.6 \quad E(x^2) = 6.8 \quad \text{Var}(x) = 6.44$$

$$(2) f(x) = 0.3\delta(x-1) + 0.7\delta(x+1)$$

$$E(x) = -0.4 \quad E(x^2) = 1 \quad \text{Var}(x) = 0.84$$

$$(3) f(x) = 4e^{-4}u(x)$$

$$E(x) = \frac{1}{4} \quad E(x^2) = \frac{1}{8} \quad \text{Var}(x) = \frac{1}{16}$$

$$(4) f(x) = \begin{cases} \frac{1}{10} & -5 < x < 5 \\ 0 & \text{其他} \end{cases}$$

$$E(x) = 0$$

$$E(x^2) = \frac{25}{3}$$

$$\text{Var}(x) = \frac{25}{3}$$

1.31 **特征函数唯一性。** 结论独立高斯线性组合仍为高斯。独立随机变量和的概率密度函数是各概率密度函数的卷积。特征函数的乘积。

1-31

解: 证明: $\phi_Y(v) = \prod_{i=1}^n \phi_{X_i}(v)$

$\phi_{X_i}(v) = e^{ju_i v - \frac{1}{2} \sigma_i^2 v^2}$

故 $\phi_Y(v) = e^{jv \sum_{i=1}^n u_i - \frac{1}{2} v^2 \sum_{i=1}^n \sigma_i^2}$

$\therefore Y \sim N\left(\sum_{i=1}^n u_i, \sum_{i=1}^n \sigma_i^2\right)$

$\therefore Y$ 也为高斯分布

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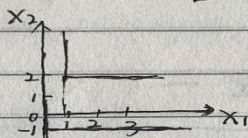
$$2.3 (1) F_X(x, \frac{1}{2}) \quad t = \frac{1}{2} \quad X(\frac{1}{2}) = \begin{cases} 0, & p = 1/2 \\ 1, & p = 1/2 \end{cases}$$

$$F_X(x, \frac{1}{2}) = \frac{1}{2} u(x) + \frac{1}{2} u(x-1)$$

$$t=1, \quad X(1) = \begin{cases} -1, & p = 1/2 \\ 2, & p = 1/2 \end{cases}$$

$$F_X(x, 1) = \frac{1}{2} u(x+1) + \frac{1}{2} u(x-2)$$

$$(2) F_X(x_1, x_2; \frac{1}{2}, 1) = \frac{1}{2} [u(x_1, x_2+1) + u(x_1-1, x_2-2)]$$



$$2.4 \quad P[B(n)=0] = 0.2 \quad P[B(n)=1] = 0.8$$

$$(1) P[\{1011\}] = 0.8 \times 0.2 \times 0.8 \times 0.8 = 0.1024$$

$$(2) E[B(n)] = 0 \times 0.2 + 1 \times 0.8 = 0.8 \quad \{B(n), B(n+1), B(n+2), B(n+3)\}$$

$$\text{平均串 } \{0.8, 0.8, 0.8, 0.8\} \quad E\{\dots\} = \{0.8, 0.8, 0.8, 0.8\}$$

$$(3) 0.8 > 0.2, 0.8^4 \text{ 最大 } \{1, 1, 1, 1\}$$

$$(2.6) X(t) = A \cos(\omega t + \theta), A, \omega \text{ 为常数 } \theta \text{ 在 } [-\pi, \pi] \text{ 均匀 } \frac{1}{2\pi} = f(\theta)$$

$$f_X(x) = \frac{f_\theta[h(x)]}{|h'(x)|} \quad f(\theta) = \frac{1}{2\pi}, \theta \in [-\pi, \pi]$$

$$h(x) = \arccos(\frac{x}{A}) - \omega t = \theta \quad 0, \text{ 其他}$$

$$= \frac{1}{2\pi} \left| \arccos(\frac{x}{A})' \right| = \frac{\frac{1}{\sqrt{1-\frac{x^2}{A^2}}}}{2\pi \sqrt{1-\frac{x^2}{A^2}}} = \frac{1}{2\pi \sqrt{1-\frac{x^2}{A^2}}}$$

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$$2.6. Y=g(x) \quad f_Y(y) = \frac{f_X[h(y)]}{|h'(y)|}, \quad a < y < b$$

$$\left. \begin{array}{l} 0, \text{ 其他} \end{array} \right\} \quad a = \min[g(-\infty), g(\infty)] \quad b = \max\{g(-\infty), g(\infty)\} \quad h(y) = g(x)$$

$$x(t) = h(\omega)(\omega t + b) \quad \theta = \arccos\left(\frac{x}{A}\right) - \omega t$$

$$f(\theta) = \frac{1}{2\pi}, \quad \theta \in [-\pi, \pi]$$

0, 其他

$$f_X(x) = f(\theta(x)) \cdot |\theta'(x)| = \frac{1}{2\pi} \left| \frac{-1}{\sqrt{1 - \frac{x^2}{A^2}}} \right| \cdot \frac{1}{A}$$

$$= \frac{1}{2\pi A \sqrt{1 - \frac{x^2}{A^2}}}, \quad |x| \leq A$$

$$2.8. f_I(i; n) = \frac{1}{2} \delta(i+1) + \frac{1}{2} \delta(i-1)$$

$$E[I_n] = \frac{1}{2}(-1) + \frac{1}{2}(1) = 0 \quad E[I_n^2] = 1$$

$$C_I(n_1, n_2) = R_I(n_1, n_2) - E[I_{n_1}] E[I_{n_2}]$$

$$C_I(n_1, n_2) = R_I(n_1, n_2) - E[I_{n_1}] E[I_{n_2}]$$

$$= R_I(n_1, n_2) = E[I_{n_1} I_{n_2}]$$

$$= \begin{cases} E[I_{n_1}] \cdot E[I_{n_2}] = 0 & n_1 \neq n_2 \\ E[I_{n_1}^2] = 1, & n_1 = n_2 \end{cases}$$

$$= \delta(n_1 - n_2)$$

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p45.2.8

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$R_{XY}(t_1, t_2) = 0$, 正交 $C_{XY}(t_1, t_2) = 0$ 无关 $F_{XY} = F_X \cdot F_Y$ 独立

$$\begin{aligned} 2.11 (1) R_{XY}(t_1, t_2) &= E[X(t_1)Y(t_2)] \\ &= E[A \sin(\omega t_1 + \theta) \cdot B \cos(\omega t_2 + \theta)] \\ &= E[AB] E[\sin(\omega t_1 + \theta) \cos(\omega t_2 + \theta)] \\ &= \frac{1}{2} E(A) E(B) E[\sin(\omega(t_1 - t_2)) + \sin(\omega(t_1 + t_2) + 2\theta)] \\ &= \frac{1}{2} E(A) E(B) \sin \omega(t_1 - t_2) \end{aligned}$$

(2) $E(A)$ 或 $E(B)$ 为 0, $X(t)$ $Y(t)$ 正交但不相关

否则, $t_1 - t_2 = \frac{k\pi}{\omega}$ k 为整数时, 正交且不相关

$X(t)$ $Y(t)$ 都为 $\sin$ 函数, 不统计独立

假设 $A = B = 1$ $X(t) = \sin(\omega t)$ $Y(t) = \cos(\omega t)$ 不独立

2.15 $X(t)$ $R_X(t_1, t_2) = 0.5 e^{-|t_1 - t_2|}$, 均值为 0。

例 1 题

$$\begin{aligned} D X &= E X^2 - (E X)^2 = R(0) - (E X)^2 \\ R(0) &= 0.5 \quad \sigma^2 = \frac{1}{2} \\ f_X(x, t) &= \frac{1}{\sqrt{2\pi} \cdot \sigma} e^{-\frac{x^2}{2\sigma^2}} = \frac{1}{\sqrt{\pi}} e^{-x^2} \end{aligned}$$

多维概率 PDF 不求求 2.18, 2.19 高维不求求!

多维: $\mu = [0]$

$$X = \begin{Bmatrix} X(t_1) \\ X(t_2) \end{Bmatrix} \quad m = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad C = \begin{pmatrix} \sigma^2 & C_X(t_1, t_2) \\ C_X(t_1, t_2) & \sigma^2 \end{pmatrix}$$

$$C_X(t_1, t_2) = E[X(t_1)X(t_2)] = R_X(t_1, t_2) = 0.5 e^{-|t_1 - t_2|}$$

$$\rho = \frac{\text{cov}(X(t_1), X(t_2))}{\sigma_X(t_1) \sigma_X(t_2)} = \frac{0.5 e^{-|t_1 - t_2|}}{0.5} = e^{-|t_1 - t_2|}$$

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P54. 2.25

$$f_{XY}(x,y) = \frac{1}{2\pi d_1 d_2 \sqrt{1-\rho^2}} \exp \left\{ -\frac{1}{2(1-\rho^2)} \left[\frac{(x-\mu_1)^2}{d_1^2} + \frac{(y-\mu_2)^2}{d_2^2} - 2 \frac{(x-\mu_1)(y-\mu_2)}{d_1 d_2} \right] \right\}$$

$$f_X(x_1, x_2; t_1, t_2) = \frac{1}{n \sqrt{1 - e^{-2|t_1 - t_2|}}} \exp \left[-\frac{x_1^2 + x_2^2 - 2x_1 x_2 e^{-|t_1 - t_2|}}{1 - e^{-2|t_1 - t_2|}} \right]$$

2.21. 予解 16进制 0~15 进制 ABCDEF 10~15

① $X(n) \in \{0, 1, \dots, 255\}$ $2^8 = 256$

② $P\{X(n) = 0x55, X(n+1) = 0x60\} = \left(\frac{1}{256}\right)^2$

③ $P\{X(n) = i\} = 1/256, i = 0, 1, \dots, 255$

$P\{X(n) = i, X(m) = j\} = \left(\frac{1}{256}\right)^2, i, j = 0, 1, \dots, 255$

④ $E[X(n)] = 0 \cdot \frac{1}{256} + 1 \cdot \frac{1}{256} + \dots + \frac{255}{256} = \frac{(255+0) \times \frac{256}{2}}{256} = \frac{255}{2}$

$D[X(n)] = E\left[\left(X(n) - \frac{255}{2}\right)^2\right] = \frac{21845}{4} ?$

$C_X(n, m) = \frac{21845}{4} \delta(n-m)$

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3.1 $u=0, \sigma^2=2$

$$f(u;t) = \frac{1}{\sqrt{2\pi} \cdot \sigma} e^{-\frac{u^2}{2\sigma^2}} = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{4}}$$

$$f(u_1, u_2; t_1, t_2) = f(u_1; t_1) f(u_2; t_2) = \frac{1}{4\pi} e^{-\frac{u_1^2 + u_2^2}{4}}$$

$$f(u_1, u_2, \dots, u_k; t_1, t_2, \dots, t_k) = \frac{1}{(2\sqrt{\pi})^k} e^{-\frac{\sum_{i=1}^k u_i^2}{4}} = \frac{1}{(2\sqrt{\pi})^k} e^{-\frac{\sum_{i=1}^k u_i^2}{4}}$$

(2). f 与 t 无关, 平稳平稳

3.3. (1) $m_x(t) = \frac{1}{3} [1 + \sin t + \cos t]$

$$R_x(t_1, t_2) = E[x(t_1)x(t_2)] = \frac{1}{3} [1 + \sin t_1 \sin t_2 + \cos t_1 \cos t_2]$$

$$= \frac{1}{3} [1 + \cos(t_1 - t_2)]$$

均值为常数, $R_x(t_1, t_2) = \frac{1}{3} [1 + \cos(t_1 - t_2)]$
非平稳 $R_x(t_1, t_2; \tau)$

3.5. $Y(t) = X^2(t)$ $X(t) = 100 \sin(\omega_0 t + \theta)$

(1). $E[X(t)] = E[100 \sin(\omega_0 t + \theta)] = 100 E[\frac{1}{2}(1 - \cos(2\omega_0 t + 2\theta))]$
 $= 50 - \frac{1}{2} E[\cos(2\omega_0 t + 2\theta)] = 50$ $\theta \in [-\pi, \pi]$ $\frac{1}{2\pi} = f(0)$

(2) $R_Y(t_1, t_2) = E[100 \sin^2(\omega_0 t_1 + \theta) \cdot 100 \sin^2(\omega_0 t_2 + \theta)]$

$$= 2500 E[(1 - \cos(2\omega_0 t_1 + 2\theta))(1 - \cos(2\omega_0 t_2 + 2\theta))]$$

$E[Y(t)]$ 为常数 $R_Y(t)$ 与 t 有关

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$$3.9 \quad R_X(\tau) = e^{-|\tau|} \quad R_Y(\tau) = \cos 2\pi\tau$$

$$(1) \quad Z(t) = X(t) + Y(t)$$

$$R_Z(t, t+\tau) = E[(X(t) + Y(t))(X(t+\tau) + Y(t+\tau))]$$

$$= E[X(t)X(t+\tau) + X(t)Y(t+\tau) + Y(t)X(t+\tau) + Y(t)Y(t+\tau)]$$

$$= R_X(\tau) + R_Y(\tau) = e^{-|\tau|} + \cos 2\pi\tau \quad \text{因为 } X, Y \text{ 统计独立}$$

$$(2) \quad R_W(t, t+\tau) = R_X(\tau) + R_Y(\tau) \quad \text{结果一致}$$

$$(3) \quad R_{ZW}(\tau) = E[Z(t) \cdot W(t+\tau)] = E[(X(t) + Y(t))(X(t+\tau) - Y(t+\tau))]$$

$$= E[X(t)X(t+\tau) - X(t)Y(t+\tau) + Y(t)X(t+\tau) - Y(t)Y(t+\tau)]$$

$$= R_X(\tau) - R_Y(\tau)$$

$\Rightarrow$

$$3.19 \quad (1) \quad S_X(\omega) = \frac{\omega^2}{\omega^4 + 3\omega^2 + 2} = \frac{2}{\omega^2 + 2} + \frac{-1}{\omega^2 + 1}$$

$$\star e^{-\alpha|t|} \leftrightarrow \frac{2\alpha}{\omega^2 + \alpha^2}, \alpha > 0$$

$$R_X(\tau) = F^{-1}\left[\frac{2 \cdot \sqrt{2}}{\omega^2 + (\sqrt{2})^2} \cdot \frac{\sqrt{2}}{2} + \frac{2}{\omega^2 + 1} \cdot \left(-\frac{1}{2}\right)\right]$$

$$= \frac{\sqrt{2}}{2} e^{-\sqrt{2}|\tau|} - \frac{1}{2} e^{-|\tau|}$$

$$EX^2 = R(0) \quad DX^2 = EX^2 - (EX)^2$$

$$R_X(0) = \frac{\sqrt{2}-1}{2}$$

$$R(0) = EX^2$$

$$e^{-\alpha|t|} \leftrightarrow \frac{2\alpha}{\omega^2 + \alpha^2}$$

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$$(2) S_X(\omega) = \begin{cases} 8\delta(\omega) + 20(1 - \frac{|\omega|}{10}) & |\omega| \leq 10 \\ 0 & |\omega| > 10 \end{cases}$$

$$R_X(\tau) = \frac{8}{2\pi} + 20 \cdot \frac{5\text{Sa}^2(5\tau)}{\pi} = \frac{4}{\pi} + \frac{100\text{Sa}^2(5\tau)}{\pi}$$

$$R_X(0) = \frac{4+100}{\pi} = \frac{104}{\pi}$$

$$(3) S_X(\omega) = p\omega^4\omega\tau_0, p \text{ 为常数}$$

$$= p \left[\frac{1 + \cos 2\omega\tau_0}{2} \right]^2$$

$$= \frac{p}{4} [1 + 2\cos 2\omega\tau_0 + \cos^2 2\omega\tau_0]$$

$$= \frac{p}{4} + \frac{p}{2} \cos 2\omega\tau_0 + \frac{p}{4} \left[\frac{1 + \cos 4\omega\tau_0}{2} \right]$$

$$= \frac{p}{4} + \frac{p}{2} \cos 2\omega\tau_0 + \frac{p}{8} + \frac{p}{8} \cos 4\omega\tau_0$$

$$= \frac{3}{8}p + \frac{p}{2} \cos 2\omega\tau_0 + \frac{p}{8} \cos 4\omega\tau_0$$

$$\cos 2\omega\tau_0 \rightarrow \frac{1}{2} [\delta(t+2\tau_0) + \delta(t-2\tau_0)]$$

$$\Rightarrow R_X(\tau) = \frac{3p}{8} \delta(\tau) + \frac{p}{4} [\delta(t+2\tau_0) + \delta(t-2\tau_0)] + \frac{p}{16} [\delta(t+4\tau_0) + \delta(t-4\tau_0)]$$

$$R_X(0) = \frac{3p}{8}$$

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3.20 ① $R(\tau) = e^{-3|\tau|}$ $S_X(\omega) = \frac{6}{\omega^2 + 9}$
 $e^{-a|\tau|} \leftrightarrow \frac{2a}{\omega^2 + a^2}, a > 0$

② $R(\tau) = 5e^{-\tau^2/8}$
 $e^{-(\frac{\tau}{a})^2} \leftrightarrow \sqrt{\pi} a e^{-(\frac{\omega a}{2})^2}$
 $a = 2\sqrt{2}$
 $S_X(\omega) = 5 \cdot \sqrt{\pi} \cdot 2\sqrt{2} e^{-(\frac{\omega \cdot 2\sqrt{2}}{2})^2} = 10\sqrt{2\pi} e^{-2\omega^2}$

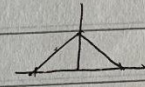
③ $R(\tau) = 4e^{-|\tau|} \cos 3\pi\tau + \cos 3\pi\tau$

$e^{-|\tau|} \rightarrow \frac{2}{1+\omega^2}$

$\cos \pi\tau \rightarrow \pi [\delta(\omega + \pi) + \delta(\omega - \pi)]$

$\cos 3\pi\tau \rightarrow \pi [\delta(\omega + 3\pi) + \delta(\omega - 3\pi)]$

$S_X(\omega) = 4 \frac{2}{1+\omega^2} \times \frac{\pi}{2\pi} [\delta(\omega + \pi) + \delta(\omega - \pi)] + \pi [\delta(\omega + 3\pi) + \delta(\omega - 3\pi)]$
 $= \frac{4\pi}{1+\omega^2} [\delta(\omega + \pi) + \delta(\omega - \pi)] + \pi [\delta(\omega + 3\pi) + \delta(\omega - 3\pi)]$

④ $R(\tau) = \begin{cases} 1 - |\tau|, & |\tau| \leq 1 \\ 0, & |\tau| > 1 \end{cases}$ 

$\text{tri}(\frac{\tau}{a}) \rightarrow \tau a^2 \frac{\omega^2}{2}, \tau > 0$

$R(\tau) \leftrightarrow S_X(\omega) = S_a(\frac{\omega}{2}) = \frac{\sin^2(\frac{\omega}{2})}{(\frac{\omega}{2})^2}$

家	长	($\frac{\omega}{2}$) ²	分	数
签	字			

第三章作业



Wensheng Zhang

$$\begin{aligned}
 3.16 \quad Z(t) &= A X(t) Y(t) \quad R_X(\tau) = 2e^{-|\tau|} \cos \omega_0 \tau \quad R_Y(\tau) = q + \exp(-\tau^2) \\
 E X^2 &= R(0) \quad (E X)^2 = R(\infty) \quad b(X) = R(0) - R(\infty) \\
 (E X)^2 &= R(\infty) = 0 \\
 (E Y)^2 &= R(\infty) = q \quad m_Y = \pm 3 \\
 \therefore E[Z(t)] &= 0 = E[A] E[X(t)] E[Y(t)] = 0 \\
 D[Z(t)] &= D[A X(t) Y(t)] = E[(A X(t) Y(t))^2] - [E[A X(t) Y(t)]]^2 \\
 &= E[A^2] E[X^2(t)] E[Y^2(t)] \\
 E A &= 2 \quad b A = q \quad E A^2 = D(A) + (E A)^2 = 13 \\
 E[X^2(t)] &= D X + (E X)^2 = R(0) = 2 \\
 E[Y^2] &= 10 \\
 \therefore D[Z(t)] &= 13 \times 2 \times 10 = 260 \\
 R_Z(t, t+\tau) &= E[Z(t) Z(t+\tau)] \\
 &= E[A X(t) Y(t) A X(t+\tau) Y(t+\tau)] \\
 &= E[A^2] R_X(\tau) R_Y(\tau) \\
 &= 26 e^{-|\tau|} (q + \exp(-\tau^2)) \cos \omega_0 \tau
 \end{aligned}$$

第三章作业



Wensheng Zhang

3.2 实信号功率谱为实的非负偶函数

① $\left(\frac{\sin w}{w}\right)^2$ ✓

② $\frac{w^2}{w^6+3w^2+3}$ ✓

③ $\frac{w^2}{w^4+1}$ ✗

④ $\frac{w^4}{jw^6+w^2+1}$ 非实

⑤ $\frac{|w|}{w^4+2w^2+1}$ ✓

⑥ $e^{-(w-1)^2}$ 非偶

3.23 $X(t)$ 平稳 $Y(t) = X(t+T) + X(t)$

$$S_Y(w) = 2S_X(w)(1 + \cos wT)$$

$$R_Y(\tau) = E[X(t+T) + X(t)][X(t+T+\tau) + X(t+\tau)]$$

$$= E[X(t+T)X(t+T+\tau) + X(t)X(t+\tau) + X(t+T)X(t+\tau) + X(t)X(t+T)]$$

$$= R_X(\tau) + R_X(\tau) + R_X(\tau+T) + R_X(\tau-T)$$

$$S_Y(w) = F[R_Y(\tau)] = 2S_X(w) + S_X(w)[e^{jwT} + e^{-jwT}]$$

$$= 2S_X(w)(1 + \cos wT)$$

3.31. $Y(t) = \sum_{n=-\infty}^{\infty} a_n g_T(t-nT-P)$

$$R_d(k) = \frac{1}{4} + \frac{1}{4}\delta(k)$$

$$G(jw) = \frac{2}{w} \sin \frac{wT}{2} = T \text{sinc} \left(\frac{wT}{2} \right)$$

$$S_Y(w) = \frac{1}{T} \times \frac{1}{4} \times \frac{4}{w^2} \sin^2 \frac{wT}{2} + \frac{2T}{T^2} \times \frac{1}{4} \times T^2 \delta(w)$$

$$= \frac{1}{w^2} \sin^2 \frac{wT}{2} + \frac{1}{2} \delta(w)$$

家	长	分	数
字			

作业 1118

1. 若实平稳随机过程 $X(t)$ 的功率谱为 $S_X(\omega)$, 平均功率为 P_X 。定义 $Y(t) = aX(t) \cos \omega_0 t$, 求 $Y(t)$ 的功率谱密度和平均功率。(设 $X(t)$ 为低通信号, 带宽远小于 ω_0)

Handwritten derivation showing the power spectrum and average power of $Y(t)$:

$$\begin{aligned} X(t), \text{ WSS}, S_X(\omega), R_X(\tau) \\ P_X &= R_X(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_X(\omega) d\omega \\ Y(t) &= a X(t) \cos \omega_0 t \\ R_Y(t, t+\tau) &= E[Y(t), Y(t+\tau)] = a^2 \cos \omega_0 t \cos \omega_0 (t+\tau) E[X(t), X(t+\tau)] \\ &= \frac{1}{2} a^2 R_X(\tau) [\cos(2\omega_0 t + \omega_0 \tau) + \cos \omega_0 \tau] \\ \langle R_Y(t, t+\tau) \rangle &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T R_Y(t, t+\tau) dt \\ &= \frac{1}{2} a^2 R_X(\tau) \cos \omega_0 \tau \end{aligned}$$

1. 若实平稳随机过程 $X(t)$ 的功率谱为 $S_X(\omega)$ ，平均功率为 P_X 。定义 $Y(t) = aX(t) \cos \omega_0 t$ ，求 $Y(t)$ 的功率谱密度和平均功率。（设 $X(t)$ 为低通信号，带宽远小于 ω_0 ）

$$\begin{aligned} \therefore R_Y(\omega) &= \mathcal{F} \{ \langle R_Y(t_1 + \tau, t_1) \rangle \} \\ &= \frac{1}{4} a^2 [S_X(\omega - \omega_0) + S_X(\omega + \omega_0)] \\ \therefore P_Y &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_Y(\omega) d\omega \\ &= \frac{1}{4} a^2 \cdot \frac{1}{2\pi} \left(\int_{-\infty}^{\infty} S_X(\omega - \omega_0) d\omega + \int_{-\infty}^{\infty} S_X(\omega + \omega_0) d\omega \right) \\ &= \frac{1}{4} a^2 \cdot 2 \cdot P_X = \frac{1}{2} a^2 \cdot P_X \end{aligned}$$

2. 已知零均值高斯平稳随机信号 $X(t)$ 的功率谱密度为 $S_X(\omega) = \frac{N_0}{2}$, $|\omega| \leq B$.

1) 计算 $X(t)$ 的平均功率; 2) 计算 $X(t)$ 的一维概率密度函数;

3) 计算自相关函数 $R_X(\tau)$ 和相关系数;

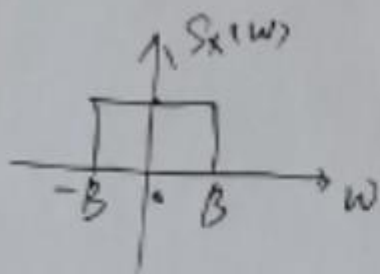
4) 写

5) 计

6) 当

2. 零均值, 高斯, 平稳

$$S_X(\omega) = \frac{N_0}{2} \quad |\omega| \leq B$$



$$\begin{aligned} 1) P_X &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_X(\omega) d\omega = \frac{1}{2\pi} \int_{-B}^B \frac{N_0}{2} d\omega \\ &= \frac{N_0 B}{2\pi} \end{aligned}$$

2) 高斯分布 $E[X(t)] = m_X = 0$

$$\sigma_X^2 = E[X^2(t)] = R_X(0) = P_X = \frac{N_0 B}{2\pi}$$

$$\therefore f_X(x) = \frac{1}{\sqrt{N_0 B}} e^{-\frac{\pi x^2}{N_0 B}} \quad -\infty < x < \infty$$

原因。

$$\begin{aligned}
 3) \quad R_X(\tau) &= \mathcal{F}^{-1}(S_X(\omega)) \\
 &= \frac{1}{2\pi} \int_{-B}^B \frac{N_0}{2} e^{j\omega\tau} d\omega \\
 &= \frac{BN_0}{2\pi} S_a(B\tau)
 \end{aligned}$$

$$\rho_X(\tau) = \frac{C_X(t, t+\tau)}{\sigma_X(t) \cdot \sigma_X(t+\tau)} = \frac{R_X(\tau)}{\sigma_X^2} = S_a(B\tau)$$

$$\begin{aligned}
 4) \quad C &= \begin{pmatrix} E(X(t_1)X(t_1)) & E(X(t_1)X(t_2)) \\ E(X(t_2)X(t_1)) & E(X(t_2)X(t_2)) \end{pmatrix} = \begin{pmatrix} \sigma_X^2 & R_X(\tau) \\ R_X(\tau) & \sigma_X^2 \end{pmatrix} \\
 \tau &= t_2 - t_1
 \end{aligned}$$

$$5) R_x(\tau) = \frac{N \cdot B}{2\pi} S_q(B\tau)$$

$$\text{当 } B\tau = k\pi \text{ 时 } \tau = \frac{k\pi}{B}, \quad k = \pm 1, \pm 2, \dots, \quad R_x(\tau) = 0.$$

$$6) E(x(t)) = 0. \quad \text{正态分布}$$

$$\text{当 } R_x(\tau) = 0 \text{ 时, } E(x(t), x(t+\tau)) = 0.$$

正交, 不相关, 统计独立

3. 已知四进制随机序列 $X(n)$ ，取值为 0, 1, 2, 3 的概率相等，且两个不同时刻的随机变量不相关。计算均值和自相关函数，是平稳随机过程吗？

$$\begin{aligned}
 3. \quad X(n), \quad P\{X(n)=i\} &= 1/4, \quad i=0, 1, 2, 3 \\
 E[X(n)] &= \sum_{i=0}^3 i \cdot P\{X(n)=i\} = 1.5 = \frac{3}{2} \\
 E[X^2(n)] &= \sum_{i=0}^3 i^2 \cdot P\{X(n)=i\} = 3.5 = \frac{14}{4} \\
 R_X(n_1, n_2) &= E[X(n_1)X(n_2)] = \begin{cases} E[X^2(n)] & n_1 = n_2 \\ E[X(n_1)] \cdot E[X(n_2)] & n_1 \neq n_2 \end{cases} \\
 &= \begin{cases} \frac{14}{4} & (n_1 = n_2) \\ \frac{9}{4} & (n_1 \neq n_2) \end{cases} = \frac{7}{4} \delta(n) + \frac{9}{4} = \frac{7}{4} \delta(n_2 - n_1) + \frac{9}{4} \\
 &\quad m = n_2 - n_1 \quad \text{为 WSS 过程}
 \end{aligned}$$

$$4) \quad x(t) = A \cos(\omega t + \theta) \quad m_A \dots \sigma_A^2$$

$$\Phi_{\theta}(v) = E(e^{j\theta v}) = \int_{-\infty}^{\infty} e^{j\theta v} \cdot f_{\theta}(v) dv$$

$$\Phi_{\theta}(v) = \Phi_{\theta}^*(-v), \quad \Phi_{\theta}(-v) = \Phi_{\theta}^*(v)$$

$$E[x(t)] = E(A) \cdot E[\cos(\omega t + \theta)]$$

$$= E(A) \cdot \frac{1}{2} E(e^{j\omega t} \cdot e^{j\theta} + e^{-j\omega t} \cdot e^{-j\theta})$$

$$= \frac{1}{2} E(A) \cdot (e^{j\omega t} \cdot \Phi_{\theta}(1) + e^{-j\omega t} \cdot \Phi_{\theta}(-1))$$

$$\frac{1}{2} \Phi_{\theta}(1) = 0 \text{ 时, } E[x(t)] = 0.$$

4. 正弦随机信号 $X(t) = A\cos(\omega_0 t + \theta)$, 其中随机变量 A 的均值为 m_A 、方差为 σ_A^2 , θ 服从特征函数为 $\Phi_\theta(v)$ 的某种分布, θ 与 A 统计独立。证明, 当 $\Phi_\theta(1) = \Phi_\theta(2) = 0$ 时, $X(t)$ 是广

$$\begin{aligned}
 R_X(t_1, t_2) &= E(X(t_1)X(t_2)) = \frac{1}{2} E(A^2) \cdot E(\cos(\omega t_1 + \omega t_2 + 2\theta) \\
 &\quad + \cos(\omega t_1 - \omega t_2)) \\
 &= \frac{1}{2} E(A^2) \cos(\omega t_1 - \omega t_2) + \frac{1}{2} E(A^2) \cdot E(\cos(\omega t_1 + \omega t_2 + 2\theta)) \\
 &= - + \frac{1}{2} E(A^2) \cdot \frac{1}{2} E(e^{j(\omega t_1 + \omega t_2) + j2\theta} + e^{-j(\omega t_1 + \omega t_2) - j2\theta})
 \end{aligned}$$

多 $\Phi_\theta(2) = 0$ 时.

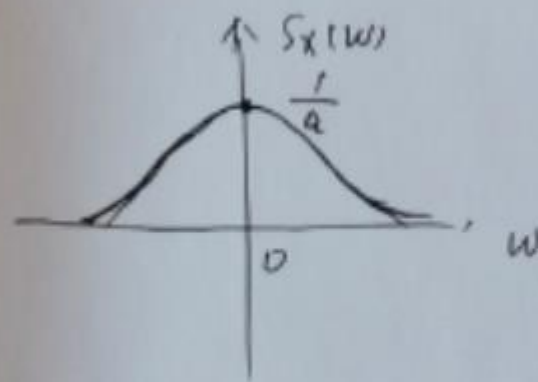
$$\begin{aligned}
 R_X(t_1, t_2) &= \frac{1}{2} E(A^2) \cos(\omega t_1 - \omega t_2) \\
 &= \frac{1}{2} (m_A^2 + \sigma_A^2) \cos(\omega t_1 - \omega t_2)
 \end{aligned}$$

5. 已知宽平稳信号的自相关函数为 $R_X(\tau) = e^{-2a|\tau|}$, 其中 $a > 0$ 为常数。计算其功率谱密度函数和平均功率; 画出功率谱密度函数的图形, 并计算其 3dB 带宽。

$$1. R_X(\tau) = e^{-2a|\tau|} \quad a > 0.$$

$$S_X(\omega) = \frac{4a}{\omega^2 + 4a^2}$$

$$P = R_X(0) = 1.$$



$$\omega = 0. \quad S_X(0) = \frac{1}{a}.$$

$$S_X(\omega_c) = \frac{1}{2} \cdot S_X(0) = \frac{1}{2a} \text{ 时.}$$

ω_c 为 3dB 带宽, 半功率点带宽

$$\therefore \frac{4a}{\omega^2 + 4a^2} = \frac{1}{2a}$$

$$\omega_c^2 = 4a^2$$

$$\therefore \omega_c = 2a \text{ (rad/s)}$$

$$2\pi f_c = \omega_c, \quad \therefore f_c = \frac{a}{\pi} \text{ (Hz)}.$$

作业 1125

- 1、什么是白噪声序列？写出其均值、方差、自相关函数、功率谱密度、相关系数。
- 2、什么是高斯白噪声序列？写出其均值、方差、自相关函数、功率谱密度、相关系数、 n 维概率密度函数。

3、 $Y(t) = X(t) \cos(\omega t + \theta)$

为
平

$$R_Y(t, t+\tau) = E[X(t)X(t+\tau)] \cdot E[\cos(\omega t + \theta) \cos(\omega t + \omega\tau + \theta)]$$

$$= \frac{1}{2} R_X(\tau) \cos \omega\tau$$

$$\therefore S_Y(\omega) = \frac{1}{4} [S_X(\omega + \omega) + S_X(\omega - \omega)]$$

$$\text{设 } P_X = R_X(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_X(\omega) d\omega$$

$$\therefore P_Y = \frac{1}{4} \times 2 \times P_X = \frac{1}{2} P_X.$$

4、已知实平稳信号 $X(t)$ 的自相关函数为 $R_X(\tau)$ ， $X(t)$ 受随机正弦信号 $Y(t) = A\cos(\omega_0 t + \theta)$ 的干扰， $X(t)$ 与 $Y(t)$ 统计独立，且 A 、 ω_0 为常数， θ 是在 $[0, 2\pi)$ 均匀分布的随机变量。受干扰后的信号为 $Z(t) = X(t) + Y(t)$ 。求： $S_Z(\omega)$ 和 $S_{XZ}(\omega)$ 。

$$4. \quad Z(t) = X(t) + Y(t), \quad R_X(t, t+\tau) = R_X(\tau) = E[X(t), X(t+\tau)]$$

$$Y(t) = A \cos(\omega_0 t + \theta)$$

$$E[Y(t)] = 0, \quad R_Y(t, t+\tau) = \frac{1}{2} A^2 \cos \omega_0 \tau.$$

$$R_Z(t, t+\tau) = E[Z(t), Z(t+\tau)] = R_X(\tau) + \frac{1}{2} A^2 \cos \omega_0 \tau = R_Z(\tau).$$

$$R_{XZ}(t, t+\tau) = E[X(t), Z(t+\tau)] = R_X(\tau).$$

5、设信号 $g(t)$ 是高度为 1 的矩形脉冲信号， $0 \leq t \leq T$ ， T 为常数。 $g(t)$ 的自相关函数定义为 $R_g(\tau) =$

1) 计算 $g(t)$

2) 证明：

3) 证明：

标。

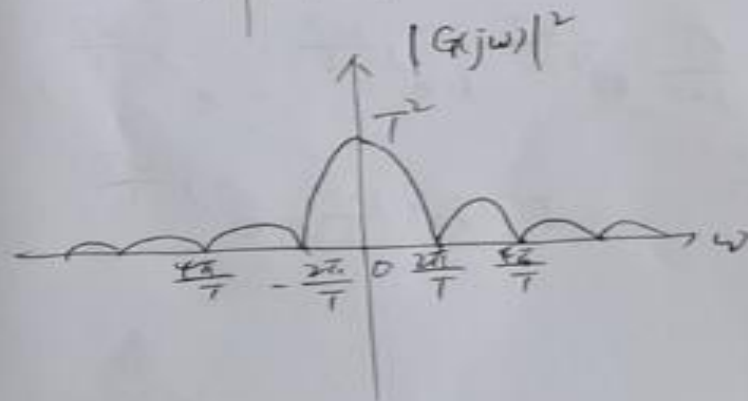
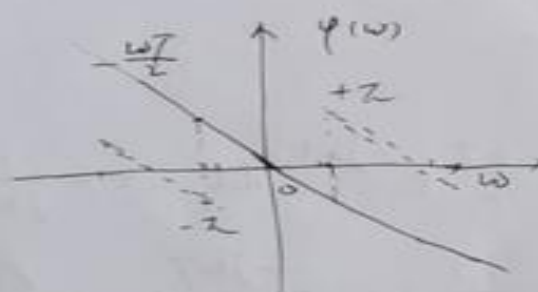
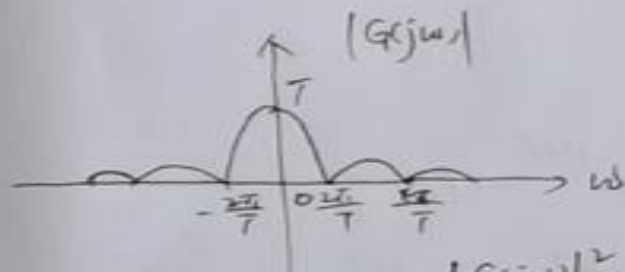
$$g(t) = \begin{cases} 1 & 0 \leq t \leq T \\ 0 & \text{其它} \end{cases}$$

$$G(j\omega) = \int_0^T e^{-j\omega t} dt = T \cdot \text{sinc}\left(\frac{\omega T}{2}\right) e^{-j\frac{\omega T}{2}}$$

$$|G(j\omega)| = T \left| \text{sinc}\left(\frac{\omega T}{2}\right) \right| \quad \varphi(\omega) = -\frac{\omega T}{2}$$

$$R_g(\tau) = g(\tau) * g(-\tau)$$

$$R_g(\omega) = G(j\omega) \cdot G^*(-j\omega) = G(j\omega) \cdot G(j\omega) = |G(j\omega)|^2$$



关键点坐

6、两个随机变量 X 、 Y ，其联合概率密度函数为 $f_{XY}(x, y)$ 。随机变量 $Z = g(Y)$ ，写出条件数学期望 $E[g(Y) | X = x]$ 的有关计算公式。根据等式 $E[g(Y)] = E[E[g(Y) | X]]$ ，写出随机变量 X 是离散和连续情况下（内容）

$$1) X, Y \quad f_{XY}(x, y)$$

$$f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)}$$

$$E(g(Y)|X=x) = \int_{-\infty}^{\infty} g(y) \cdot f_{Y|X}(y|x) dy$$

$$2) E(g(Y)) = E(E(g(Y)|X))$$

$$E(g(Y)) = \int_{-\infty}^{\infty} E(g(Y)|X=x) \cdot f_X(x) dx$$

$$E(g(Y)) = \sum_{k=0}^{\infty} E(g(Y)|X=x_k) \cdot P\{X=x_k\}$$

$$\begin{aligned} * G(j\omega) &= \int_{-T}^T e^{-j\omega t} dt = -\frac{1}{j\omega} (e^{-j\omega T} - 1) \\ &= \frac{1}{j\omega} (1 - e^{-j\omega T}) = \frac{1}{j\omega} e^{-j\frac{\omega T}{2}} (e^{j\frac{\omega T}{2}} - e^{-j\frac{\omega T}{2}}) \\ &= \frac{2}{\omega} \sin \frac{\omega T}{2} \cdot e^{-j\frac{\omega T}{2}} = T \cdot \text{sinc} \left(\frac{\omega T}{2} \right) e^{-j\frac{\omega T}{2}} \end{aligned}$$

第五章

$$5.2 \quad R_X(\tau) = A^2 + B e^{-|\tau|} \quad m_X^2 = A^2 \quad m_X = \pm A$$

$$h(t) = \pm e^{-\omega t} u(t), \quad (\omega > 0) \quad H(j\omega) = \frac{1}{\omega + j0}$$

$$m_Y = m_X \cdot \int_{-\infty}^{\infty} h(t) dt = m_X \cdot H(j\omega) \Big|_{\omega=0}$$

$$= \pm A \cdot \int_0^{\infty} t e^{-\omega t} dt = \pm \frac{A}{\omega^2}$$

第五章

$x(t)$ 为输入, $E(t)$ 为输出, $E(t) = Y(t) - X(t)$.

$$H(j\omega) = \frac{R}{R + \frac{1}{j\omega C}} = \frac{j\omega CR}{1 + j\omega CR}$$

$$S_E(\omega) = S_X(\omega) \cdot |H(j\omega)|^2 = S_X(\omega) \cdot \frac{\omega^2 R^2 C^2}{1 + \omega^2 R^2 C^2}$$

5.4

$$R_X(t, t+\tau) = E[X(t)X(t+\tau)]$$

$$= E[(X_0 + \omega(\omega t + \Phi))(X_0 + \omega(\omega t + \omega \tau + \Phi))]$$

$$= E[X_0^2] + E[X_0] \cdot E[\omega(\omega t + \omega \tau + \Phi)]$$

$$+ E[X_0] \cdot E[\omega(\omega t + \Phi)] + E\left[\frac{1}{2}\omega\omega\tau + \frac{1}{2}\omega\right. \\ \left.(\omega t + \omega \tau + 2\Phi)\right]$$

$$= \int_0^1 x^2 dx + \frac{1}{2}\omega\omega\tau$$

$$= \frac{1}{3} + \frac{1}{2}\omega\omega\tau$$

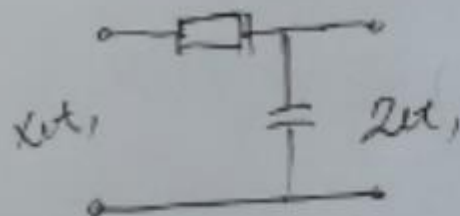
5.4

$$\therefore S_X(\omega) = \frac{2\pi}{3} \delta(\omega) + \frac{\pi}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

$$\begin{aligned} \therefore S_Y(\omega) &= S_X(\omega) \cdot |H(j\omega)|^2 & |H(j\omega)|^2 &= \frac{1}{1 + \omega^2 R^2 C^2} \\ &= \frac{2\pi}{3} \delta(\omega) + \frac{\pi}{2} \left(\frac{1}{1 + \omega_0^2 R^2 C^2} \delta(\omega - \omega_0) \right. \\ &\quad \left. + \frac{1}{1 + \omega_0^2 R^2 C^2} \delta(\omega + \omega_0) \right) \end{aligned}$$

$$\therefore R_Y(\tau) = \frac{1}{3} + \frac{1}{1 + \omega_0^2 R^2 C^2} \cdot \frac{1}{2} \cos \omega_0 \tau$$

5.5



$$H(j\omega) = \frac{1}{1 + j\omega RC}$$

$$Y(t) = Z(t)$$

$$S_X(\omega) = \frac{N_0}{2}$$

$$S_Z(\omega) = \frac{N_0}{2} \cdot |H(j\omega)|^2 = \frac{N_0}{2} \cdot \frac{1}{1 + \omega^2 R^2 C^2} \quad \frac{1}{RC} = a$$

$$= \frac{N_0}{2} \cdot \frac{\frac{1}{R^2 C^2}}{\frac{1}{R^2 C^2} + \omega^2} = \frac{N_0}{4RC} \cdot \frac{2 \cdot \frac{1}{RC}}{\frac{1}{R^2 C^2} + \omega^2}$$

$$\therefore R_Z(\tau) = \frac{N_0}{4RC} e^{-\frac{|\tau|}{RC}}$$

5.5

$$R_Y(t, t+\tau) = E[Z(t)Z(t+\tau)] = E[Z(t), Z(t), Z(t+\tau), Z(t+\tau)]$$

$Z(t)$, 高斯分布 $Z(t+\tau)$, 高斯分布

利用特性函数

$$= [R_Z(0)]^2 + 2[R_Z(\tau)]^2$$

$$= \frac{N_0^2}{16R^2C^2} + \frac{2 \cdot N_0^2}{16R^2C^2} e^{-\frac{2|\tau|}{RC}}$$

$$= R_Y(\tau)$$

$$\therefore S_Y(\omega) = \frac{N_0^2}{16R^2C^2} (2\pi\delta(\omega) + 2 \cdot \frac{\frac{2}{RC}}{\omega^2 + \frac{4}{R^2C^2}})$$

$$= \frac{N_0^2}{16R^2C^2} (2\pi\delta(\omega) + \frac{8RC}{4 + \omega^2R^2C^2})$$

$$= \frac{N_0^2}{8R^2C^2} (\pi\delta(\omega) + \frac{4RC}{4 + \omega^2R^2C^2})$$

第一章 概率论基础

- 1、随机变量的概率分布

PMF、CDF、PDF

联合分布*、边缘分布*、条件分布*

- 2、随机变量的数字特征*
- 3、随机变量的特征函数*
- 4、随机变量的正交*、不相关*、统计独立*
- 5、常用的概率分布的概率分布、数字特征和特征函数
0-1分布、均匀分布、泊松分布、指数分布、高斯分布*

第二章 随机信号

- 1、随机信号的概率特性和数字特征
 - 一维、二维的CDF*、PDF*
 - 均值*、相关函数*、协方差函数*
- 2、典型信号的基本特性
 - 1) 随机正弦信号*
 - 2) 贝努利随机序列*
- 3、多维高斯分布和高斯随机信号*
 - 多维概率密度函数、特征函数、性质

第三、四章 平稳性、功率谱密度和各态历经性

- 1、严格平稳和宽平稳信号的定义和特点*
- 2、宽平稳信号的相关函数的特点
- 3、信号的能量和功率的计算*
- 4、随机信号的功率谱密度的定义、特点
- 5、宽平稳随机信号的功率谱密度和相关函数的关系*
- 6、白噪声信号的特点*
- 7、宽平稳随机信号的各态历经性的定义和意义

第五章 随机信号通过线性系统

- 1、平稳随机信号通过线性系统的输出信号的特征*
- 2、白噪声通过线性系统*
- 3、低通白噪声、带通白噪声的基本特征
- 4、随机信号的等效噪声带宽、第一零点带宽